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Experiment No. 1

Introduction to apparatus used in Chemistry laboratory

In chemistry laboratory students use different apparatus in various experiments. So at first, students should know the name of each apparatus and its use (Fig. 1.1).

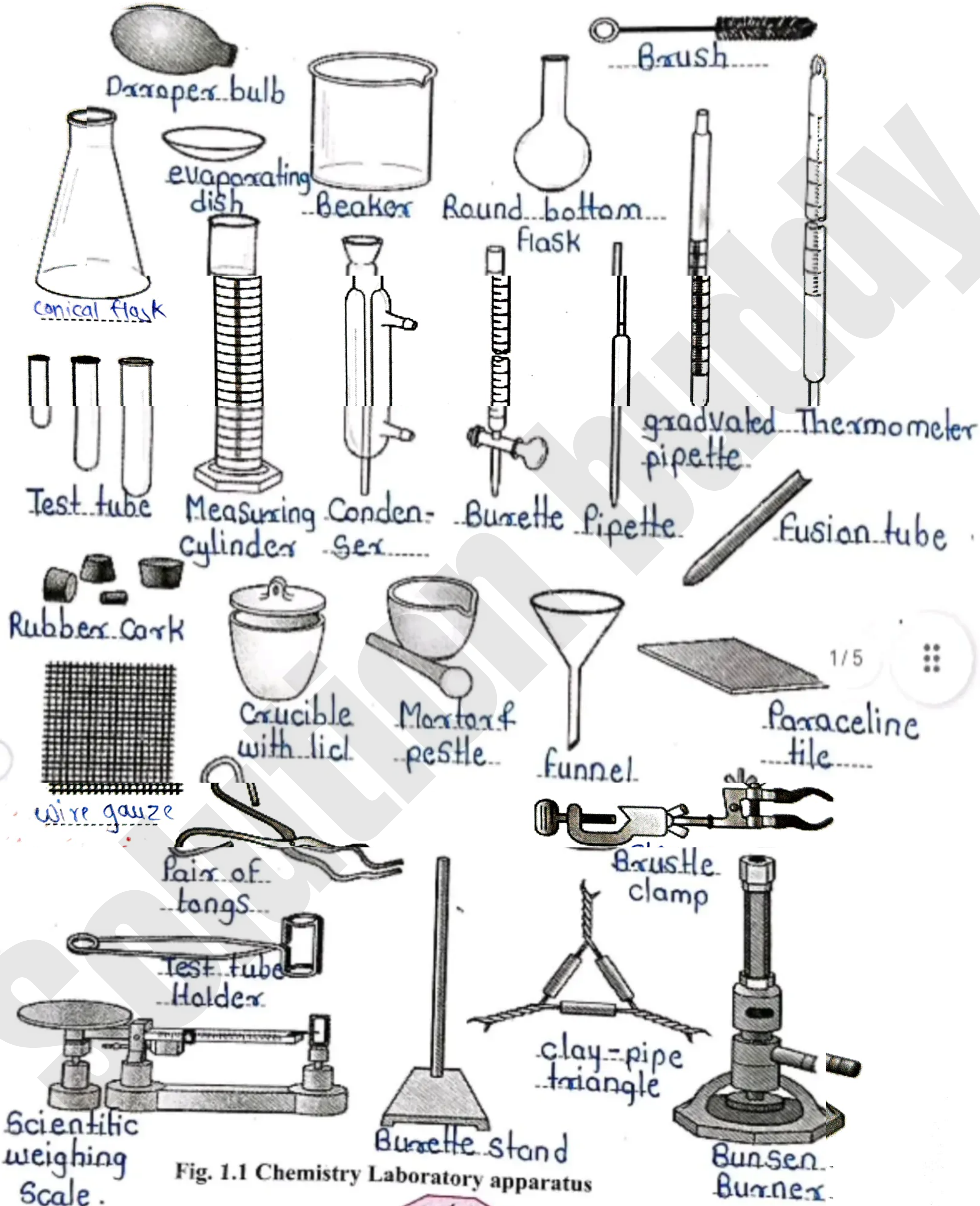

Fig. 1.1 Chemistry Laboratory apparatus

The burette: It is a long glass tube of uniform bore with a stopcock at the bottom. It is used to deliver a solution for titration. The flow of the solution for titration can be regulated or stopped as desired by means of the stopcock [Fig. 1.5(a)] Sometimes a rubber tube with a pinch cock is attached to the burette instead of the stopcock [Fig. 1.5 (b)].

How to use burette: The burette is commonly designed to hold 25 mL, 50 mL or 100 mL of the solution and is graduated in tenths of a mL. The zero mark is always at the top and the final mark (25, 50 or 100 mL) at the bottom of the burette.

Wash the burette atleast twice with water before use. Then rinse it with the solution (as does in the case of the pipette). Fill it with the solution sufficiently above the zero mark. Turn the stopcock or press the pinch cock to allow the solution to run down and fill the nozzle completely. See that no air bubble is left either in the nozzle or burette. Adjust the level of the solution at the zero mark. The lower meniscus of the solution must be in line with your eye level and the zero mark. Always take solution upto the zero mark as the starting point for every titration. (Although it is not necessary to adjust the burette each time to a zero mark this process avoids the subtraction of two volumes and hence the possibility of an error.)

Simple method of getting a correct burette reading is to hold a plain white paper behind the burette. Keep your eye level parallel with the lower meniscus. It is obvious that such a reading is correct (Fig 1.6, b) while those taken from positions (a and c) other than this, are incorrect (Fig.1.6 a).

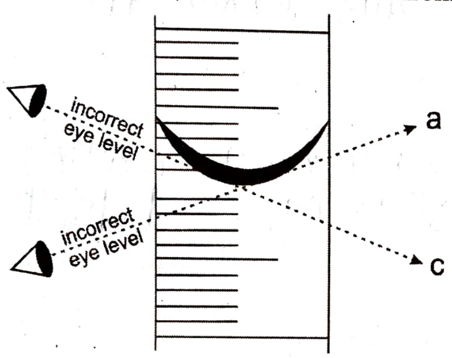

Fig 1.6 (a) Incorrect burette reading

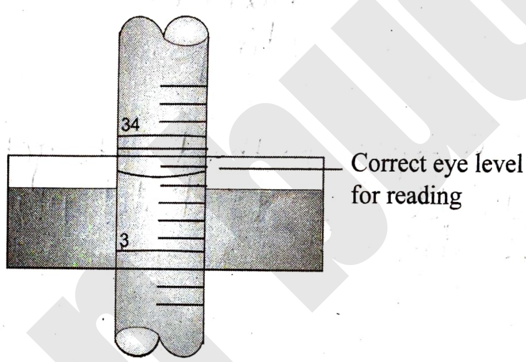

Fig. 1.6 (b) Correct burette reading

MCQ

Select the correct answer for the following multiple choice type of questions.

1. Glass vessel with a flat bottom and a long neck on which a mark is etched to show the level is called ....
   - a. flat bottom flask
   - b. volumetric flask
   - c. conical flask
   - d. beaker
2. Least count in 100 mL measuring cylinder is ....
   - a. 10 mL
   - b. 2 mL
   - c. 1 mL
   - d. 0.1 mL
3. The correct burette reading in following diagram is ....

   - a. 19.5 mL
   - b. 19.6 mL
   - c. 19.7 mL
   - d. 19.8 mL

4. What can you do, when last drop of the solution remain in the jet / nozzle of the pipette?
   a. Must blow out this drop
   b. Touch the lower end of pipette to conical flask
   c. Equal amount of solution is added to solution
   d. Remove the drop of solution at any cost.
5. During the observation of burette reading, eye level and solution level in the burette must be in
   a. same level b. slightly upside level
   c. slightly below the level d. middle level

Short answer questions

1. Why it is necessary to rinse the pipette after washing with water?

It is necessary to rinse the pipette with the solution to be used because water droplets remaining inside the pipette may dilute the solution and cause errors in measurement.

2. What will happen if we do not remove the air bubble from nozzle of burette?

If air bubbles are present in the nozzle of the burette, they occupy space and cause an incorrect volume reading, resulting in experimental errors.

3. What is the simple method of getting a correct reading of burette?

A correct burette reading can be obtained by keeping the eye at the level of the lower meniscus and taking the reading against a plain white background to avoid parallax error.

4. Why it is necessary to rinse the burette after washing with water?

It is necessary to rinse the burette with the solution to be used because any water remaining inside the burette may dilute the solution and affect the accuracy of the titration.

5. Always take the zero mark of lower meniscus as the starting point for every titration. Give reason.

The zero mark of the lower meniscus is taken as the starting point to avoid errors in volume measurement and to obtain accurate and convenient readings during titration.

6. Observe the chemistry laboratory apparatus on page no 4. (Fig 1.1) write their names in space provided.

Ans :- The apparatus used in volumetric analysis are burette, pipette, conical flask, beaker, funnel, burette stand and clamp.

Remark and sign of teacher: